

Nonlinear adaptive control for quadrotor flying vehicle

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Abstract In this work, we deal with autonomous tracking and disturbance rejection problem of quadrotor vehicle flying in uncertain environment. The vehicles kinematic and modeling error uncertainties are associated with external disturbance, inertia, mass, and nonlinear aerodynamic forces and moments. The proposed method integrate the techniques from adaptive control and robust control theory. Robust and adaptive control algorithms for translational and orientation tracking are derived using Lyapunov method. It is shown in our analysis that the altitude, position, and attitude tracking errors are bounded and their bounds asymptotically converge to zero in Lyapunov sense.

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Simulation results on a commercial quadrotor flying vehicle are given to demonstrate the effectiveness of theoretical arguments for real world application.

Keywords Robust and adaptive control · Quadrotor UAV · Lyapunov Method

1 Introduction

Last decade, extensive research on small scale four rotor unmanned aerial vehicles (UAV) has been carried out by many researchers and industrials all over the world. The interest of such small scale vehicles are growing in military and civilian applications, such as, surveillance, inspection, search, and rescue mission in dangerous or awkward environments that are inaccessible for human intervention. In contrast with single rotor helicopter, quadrotor UAV has many advantages, such as, low cost, hovering capability, vertical take-off and vertical-landing ability, small size, noiseless operation, and easy maintenance. However, accurate autonomous flight tracking system design for small scale quadrotor capability of flying in uncertain environment against uncertainty is challenging because of its underactuated property, coupling between translational and rotational dynamics, inherent nonlinearity associated with the dynamical model. There are many different control schemes available in the literature to deal with the above mentioned difficulties associated with the UAV system. Authors in [1–3] proposed the proportional-

integral-derivative (PID) and linear quadratic regulator. These algorithms were developed to achieve an autonomous hovering flight control for UAV systems. However, these linear designs for simple hovering flight are not robust and exhibiting poor tracking performance in the presence of modeling errors. To improve the hovering performance, popular backstepping control techniques have been employed to address the problem of coupling in the pitch-yaw-roll and the problem of coupling in kinematics and dynamic underactuated UAV system [4–6]. In view of this control structure, one can notice that the design and stabilization of the hovering flight control system were difficult and complicated. To simplify this design, one may include integral action with the backstepping technique so-called integral backstepping algorithm [7–9]. The idea of including PID term with classical backstepping design was to reduce the steady state tracking errors while maintain asymptotic stability of the whole closed loop system. Authors in [10] proposed model based dynamic inversion mechanism for the hovering flight of the quadrotor control system design. Their design combines tracking controller with a dynamic inversion inner loop with an internal dynamics stabilization outer loop. Authors in [11] introduced robust optimal attitude controller for multi-rotor helicopters. The design combined nominal optimal controller with robust compensator. The nominal controller was derived using linear quadratic regulation while the robust compensator was used to deal with uncertainties. Recent research efforts have been focused on adaptive design to deal with uncertainty associated with the dynamics of the UAV system. Results in this direction can be found in [12–15]. In [15], authors proposed sliding mode design which ensures the stability of the roll and pitch angle. The design can be used to regulate the tracking errors of the position-yaw angle in Cartesian coordinate to zero. Nonlinear robust control law was proposed in [14] for quadrotor UAV in the presence of dynamic uncertainty. The authors showed that the tracking errors of the position and yaw rotation were uniformly ultimately bounded via using Lyapunov-like energy function. In [14], authors combined robust H_∞ tracking controller with backstepping controller to control uncertain UAV system. Backstepping controller design was employed to track the desired trajectory while nonlinear robust H_∞ controller used to stabilize rotational dynamics of the UAV. In [12, 13], nonlinear adaptive control using backstepping technique was proposed for underactu-

ated quadrotor system in the presence of model parameter uncertainty. Their method can be used to ensure the boundedness of the tracking errors of the position and yaw rotation via using Lyapunov function. However, algorithm design, evaluation procedure and closed loop stability analysis is very complicated as the method used many steps and various augmented and auxiliary signals. In view of existing reported results, one can also notice that the vehicle model and control designs are free from external disturbance uncertainty. On the other hand, these designs were based on simplified model without considering large modeling error uncertainties appearing from the variation of large payload, aerodynamic friction and gyroscopic torques. Moreover, unpredictable changes in uncertain indoor and outdoor environment may also increase the modeling errors uncertainty significantly making the flight control system design even more complicated. In this paper, we propose nonlinear adaptive flight control strategy for four rotor UAV capability of flying in the presence of uncertainties associated with uncertain environment, payload mass, inertia matrix, actuator faults, and nonlinear aerodynamic damping force and moment. Algorithms for altitude, position, and attitude tracking design combine auxiliary error terms with adaptation laws to deal with the flight dynamic changes as a result of large modeling errors and external disturbance uncertainties. The proposed design is divided into two loops as internal and external loop by assuming that the outer loop runs slower than the inner loop. The internal loop is constructed using attitude control dynamics. Attitude controller generates roll, pitch, and yaw control moment. The outer loop combines position controller in x and y axis. The position controller provides desired tracking trajectory for the roll and pitch while the desired trajectory for yaw comes directly from the designer. Altitude controller outputs the desired thruster for vertical take-off and landing of the vehicle. The algorithms are developed through Lyapunov-like energy functional. It has been shown from our convergence analysis that tracking errors of the altitude, position, and orientation are bounded and their bounds converge to zero asymptotically in the Lyapunov sense. These arguments are demonstrated through simulation and experimental demonstration on a commercial quadrotor flying vehicle in the presence of large modeling errors and disturbance uncertainties.

This paper is organized as follows. In Sect. 2, kinematics and dynamics of the four rotor flying vehi-

cles are given. Nonlinear adaptive tracking systems are developed in Sect. 3. A detail stability analysis is also presented in Sect. 3. Simulation results and experimental demonstrations are presented in Sect. 4. Finally, conclusion is given in Sect. 5.

2 Model dynamics

In this section, we first model unmanned four rotor flying robot vehicles as depicted in Fig. 1. To derive quadrotor motion dynamics, let us consider two main reference frames as earth fixed inertial reference Ξ defined by (E_{sa}, E_{sb}, E_{sc}) and body-fixed frame Σ defined by (e_{sa}, e_{sb}, e_{sc}) which is attached to the center of mass of the quadrotor vehicle. Then, the position and velocity of the center of the mass of the vehicle expressed in the inertial frame Ξ is defined as $x_s = [x \ y \ z]^T \in \mathbb{R}^3$ and $v_s = [v_{s1} \ v_{s2} \ v_{s3}]$. Let us define the orientation and angular velocity of the center of the mass of the vehicle expressed in the body-fixed frame by three Euler angles as $\eta = [\phi \ \theta \ \varphi]^T$ and by the three rotational velocities as $\Omega_s = [\Omega_{s1} \ \Omega_{s2} \ \Omega_{s3}]$. The transformation of vectors from rigid body frame to the inertial frame can be derived by the rotation matrix as $R_s \in \mathbb{R}^{3 \times 3}$

$$R_s = \begin{bmatrix} C_\phi C_\varphi & S_\phi S_\theta C_\varphi - C_\phi S_\varphi & C_\phi S_\theta C_\varphi + S_\phi S_\varphi \\ C_\theta S_\varphi & S_\phi S_\theta S_\varphi + C_\phi C_\varphi & C_\phi S_\theta S_\varphi - S_\phi C_\varphi \\ -S_\phi & S_\phi C_\theta & C_\phi C_\theta \end{bmatrix}, \quad (1)$$

where C_θ and S_θ denotes $\cos\theta$ and $\sin\theta$.

Using Newton–Euler formulation in the body-fixed frame, the dynamic model for four rotor flying vehicle subjected to translational forces and control torques can be derived as

$$\dot{x}_s = v_s, \quad (2)$$

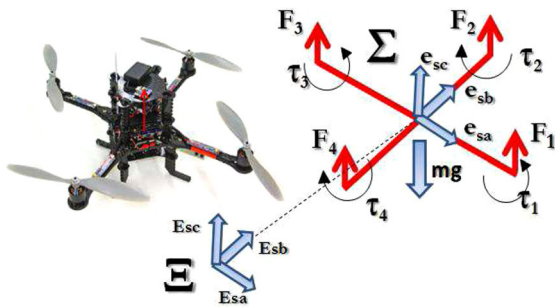


Fig. 1 Coordinate system of four rotor UAV system

$$m_s \dot{v}_s = -m_s g \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + F_t + F_d, \quad (3)$$

$$\dot{R}_s = R_s S(\Omega_s), \quad (4)$$

$$I \dot{\Omega}_s = -(\Omega_s \times I \Omega_s) + u_t + u_g + u_a, \quad (5)$$

where $m \in \mathbb{R}$ and $I \in \mathbb{R}^{3 \times 3} = \text{diag}[I_x, I_y, I_z]$ denotes the mass and symmetric positive definite constant inertia matrices and $S(\Omega_s)$ with the skew-symmetric matrix $S(\Omega_s)$ which satisfies the skew-symmetric property that can be defined as follow

$$S(\Omega_s) = \begin{bmatrix} 0 & -\Omega_{s3} & \Omega_{s2} \\ \Omega_{s3} & 0 & -\Omega_{s1} \\ -\Omega_{s2} & \Omega_{s1} & 0 \end{bmatrix}. \quad (6)$$

The angular velocity transformation matrices using the Euler angles can be used to relate the rate of change of the angular velocities in the body-fixed frame as

$$\begin{pmatrix} \Omega_{s1} \\ \Omega_{s2} \\ \Omega_{s3} \end{pmatrix} = \begin{pmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \cos\theta \sin\phi \\ 0 & -\sin\phi & \cos\phi \cos\theta \end{pmatrix} \begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\varphi} \end{pmatrix}. \quad (7)$$

Unlike exiting model, the nonlinear aerodynamic drag forces F_d is assumed to be varied with the proportional and square of the linear velocities as

$$\begin{pmatrix} F_{dx} \\ F_{dy} \\ F_{dz} \end{pmatrix} = \begin{bmatrix} -\delta_x & 0 & 0 \\ 0 & -\delta_y & 0 \\ 0 & 0 & -\delta_z \end{bmatrix}, \quad (8)$$

where $\delta_x = \zeta_1 |v_{s1}| + \zeta_2 |v_{s1}|^2$, $\delta_y = \zeta_3 |v_{s2}| + \zeta_4 |v_{s2}|^2$, $\delta_z = \zeta_5 |v_{s3}| + \zeta_6 |v_{s3}|^2$ with positive constants ζ_i with $i = 1, 2, \dots, 6$. Now, the thrust forces generated by four rotors can be derived as

$$F_t = R_s \cdot d \Sigma_{i=1}^4 \omega_i \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad (9)$$

with the trust factor $d > 0$ and ω_i is the speed of the rotors with $i = 1, 2, 3, 4$. The trust factor depends on the air density, rotor blade collective pitch and geometric blade characteristics. The torques developed in the epicenter of a quadrotor helicopter by the propellers can be defined as

$$u_t = \begin{bmatrix} dl(\omega_3^2 - \omega_1^2) \\ dl(\omega_4^2 - \omega_2^2) \\ \alpha_r(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \end{bmatrix}, \quad (10)$$

where l is the distance between the center of the mass and the rotor axes and α_r is the drag factor depending on the air density, rotor blade collective pitch and

geometric blade characteristics. The nonlinear aerodynamic friction torques u_a in our model are assumed to be varying with the combination of the linear and square of the body's angular velocities as follows

$$\begin{pmatrix} u_{ax} \\ u_{ay} \\ u_{az} \end{pmatrix} = \begin{bmatrix} -\pi_{1x} & 0 & 0 \\ 0 & -\pi_{2y} & 0 \\ 0 & 0 & -\pi_{3z} \end{bmatrix}, \quad (11)$$

where $\pi_{1x} = \xi_1|\Omega_{s1}| + \xi_2|\Omega_{s1}|^2$, $\pi_{2y} = \xi_3|\Omega_{s2}| + \xi_4|\Omega_{s2}|^2$, $\pi_{3z} = \xi_5|\Omega_{s3}| + \xi_6|\Omega_{s3}|^2$ with the positive constants of nonlinear aerodynamic coefficients $\xi_i = 1, 2, 3, 4, 5, 6$. The gyroscopic effects are produced as a result of the rigid body rotation and propeller rotation coupled with the body rotation. Therefore, the gyroscopic torques are generated by the rotors as they moves along with the rotor mast with the body-fixed frame as defined as

$$u_g = -I_r \left(\Omega_s \times \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) \cdot (\omega_1 - \omega_2 + \omega_3 - \omega_4), \quad (12)$$

where I_r is the inertia of the rotor blade. In flying vehicle, the rotational velocities of the four rotors ω_i are usually used as an input variable for designing an autonomous system. In this work, we choose the following input variables for the four rotors of the vehicle

$$\begin{aligned} u_1 &= d \left(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2 \right), \\ u_2 &= dl \left(\omega_3^2 - \omega_1^2 \right), \\ u_3 &= dl \left(\omega_4^2 - \omega_2^2 \right), \\ u_4 &= \alpha_r \left(\omega_1^2 + \omega_3^2 - \omega_2^2 - \omega_4^2 \right). \end{aligned} \quad (13)$$

Using (6)–(13), we can describe overall dynamical model of the quadrotor flying vehicles as follows

$$\begin{aligned} \ddot{x} &= \frac{1}{p_a} (\cos \phi \sin \theta \cos \varphi + \sin \phi \sin \phi) u_1 \\ &\quad - Y_{v_{s1}}(v_{s1})\theta_{vs1} + u_{dx}, \\ \ddot{y} &= \frac{1}{p_a} (\cos \phi \sin \theta \sin \varphi - \sin \phi \cos \varphi) u_1 \\ &\quad - Y_{v_{s2}}(v_{s1})\theta_{vs2} + u_{dy}, \\ \ddot{z} &= \frac{1}{p_a} (\cos \phi \cos \theta) u_1 - g - Y_{v_{s3}}(v_{s3})\theta_{vs3} + u_{dz}, \\ \ddot{\phi} &= p_b (\dot{\phi}\dot{\phi}) + p_{b1}u_2 - p_{b2}f(u)\dot{\theta} - Y_{\Omega_{s1}}(\Omega_{s1})\theta_{\Omega_{s1}} \\ &\quad + u_{rx}, \end{aligned}$$

$$\ddot{\theta} = p_c (\dot{\phi}\dot{\phi}) + p_{c1}u_3 + p_{c2}f(u)\dot{\phi} - Y_{\Omega_{s2}}(\Omega_{s2})\theta_{\Omega_{s2}} + u_{ry},$$

$$\ddot{\varphi} = p_d (\dot{\theta}\dot{\phi}) + p_{d1}u_4 - Y_{\Omega_{s3}}(\Omega_{s3})\theta_{\Omega_{s3}} + u_{rz}, \quad (14)$$

with $Y_{\Omega_{s3}}(\Omega_{s3})\theta_{\Omega_{s3}} = \frac{\pi_{1z}}{I_z}$, $u_d(t) = [u_{dx}(t), u_{dy}(t), u_{dz}(t)]^T$, $u_r(t) = [u_{rx}(t), u_{ry}(t), u_{rz}(t)]^T$, $p_d = (\frac{p_z}{I_z})$, $p_1 = (I_y - I_z)$, $p_2 = I_z - I_x$, $Y_{v_{s3}}(v_{s3}) = [v_{s3}, v_{s3}|v_{s3}|]$, $Y_{\Omega_{s1}}(\Omega_{s1}) = [\Omega_{s1}, \Omega_{s1}|\Omega_{s1}|]$, $Y_{\Omega_{s2}}(\Omega_{s2}) = [\Omega_{s2}, \Omega_{s2}|\Omega_{s2}|]$, $Y_{\Omega_{s3}}(\Omega_{s3}) = [\Omega_{s3}, \Omega_{s3}|\Omega_{s3}|]$, $Y_{v_{s3}}(v_{s3})\theta_{vs3} = \frac{\delta_z}{m}$, $p_a = m$, $p_b = (\frac{p_1}{I_x})$, $p_c = (\frac{p_2}{I_y})$, $p_3 = (I_x - I_y)$, $p_{b1} = (\frac{I}{I_x})$, $p_{c1} = (\frac{I}{I_y})$, $p_{d1} = (\frac{I}{I_z})$, $p_{b2} = (\frac{I}{I_x})$, $p_{c2} = (\frac{I}{I_y})$, $Y_{v_{s1}}(v_{s1})\theta_{vs1} = \frac{\delta_x}{m}$, $Y_{\Omega_{s1}}(\Omega_{s1})\theta_{\Omega_{s1}} = \frac{\pi_{1x}}{I_x}$, $Y_{v_{s2}}(v_{s2})\theta_{vs2} = \frac{\delta_y}{m}$, $Y_{\Omega_{s2}}(\Omega_{s2})\theta_{\Omega_{s2}} = \frac{\pi_{1y}}{I_y}$, $Y_{v_{s1}}(v_{s1}) = [v_{s1}, v_{s1}|v_{s1}|]$, $\theta_{vs1} = [\frac{\xi_1}{m}, \frac{\xi_2}{m}]^T$, $\theta_{vs2} = [\frac{\xi_3}{m}, \frac{\xi_4}{m}]^T$, $\theta_{vs3} = [\frac{\xi_5}{m}, \frac{\xi_6}{m}]^T$, $\theta_{\Omega_{s1}} = [\frac{\xi_1}{I_x}, \frac{\xi_2}{I_x}]^T$, $\theta_{\Omega_{s2}} = [\frac{\xi_3}{I_y}, \frac{\xi_4}{I_y}]^T$, $\theta_{\Omega_{s3}} = [\frac{\xi_5}{I_z}, \frac{\xi_6}{I_z}]^T$, $Y_{v_{s2}}(v_{s1}, v_{s2}) = [v_{s2}, v_{s2}|v_{s1}|]$, $Y_{v_{s3}}(v_{s3}) = [v_{s3}, v_{s3}|v_{s3}|]$, $Y_{\Omega_{s3}}(\Omega_{s3}) = [\Omega_{s3}, \Omega_{s3}|\Omega_{s3}|]$, $f(u) = (\omega_1 - \omega_2 + \omega_3 - \omega_4)$, $Y_{\Omega_{s1}}(\Omega_{s1}) = [\Omega_{s1}, \Omega_{s1}|\Omega_{s1}|]$, $Y_{v_{s2}}(v_{s1}, v_{s2}) = [v_{s2}, v_{s2}|v_{s1}|]$, $Y_{\Omega_{s2}}(\Omega_{s2}) = [\Omega_{s2}, \Omega_{s2}|\Omega_{s2}|]$. For our analysis, we consider that the terms $f(u)$ are bounded. Notice that the quadrotor system considered for our design is subjected to external disturbances $u_d(t)$ and $u_r(t)$.

Remark 1 It is assumed that the nonlinear aerodynamic forces and torques acting on the flying vehicles are varying with the combination of the linear and square of the intensity of the body's translational and orientation velocity vector. This is, in fact, realistic with the fact that the intensity of the aerodynamic forces and moments does not grow faster than the square of the velocity.

3 Algorithm design and stability analysis

In this section, we develop nonlinear adaptive tracking system that can be used for quadrotor unmanned vehicle flying in uncertain indoor and outdoor environment. Notice from model that the mass, inertia and nonlinear aerodynamic damping friction are uncertain causing modeling error uncertainty. This is due to the fact that the pay load mass and inertia may vary with the different flight mission making large modeling error causing poor tracking performance. Due to the presence of unpredictable variation in flying environment, very strong dynamic coupling and nonlinearities associated

with the translational and rotational dynamics of the vehicle makes flight control system even more complex. As a result, it is very challenging task to design autonomous flight control system for a small unmanned flying vehicles to ensure desired stability and tracking. To deal with these problems, we introduce robust adaptive tracking algorithms to learn and compensate with uncertain modeling error and disturbance uncertainties that entering into the closed loop system. The proposed method combines the techniques from adaptive control with robust control theory [16, 17]. The design objective is to develop input algorithms for altitude, attitude and position such that the altitude, position, and yaw trajectory of the vehicle tracks a time varying reference altitude, position, and yaw trajectory in the presence of modeling error and disturbance uncertainty. Throughout our design and stability analysis, we consider the following assumption.

Assumption 1 The desired position and orientation trajectories x_d , η_d , $\dot{x}_d$, $\dot{\eta}_d$, $\ddot{x}_d$ and $\ddot{\eta}_d$ are bounded.

Assumption 2 The external disturbances $u_d(t)$ and $u_r(t)$ are considered to be bounded and satisfy the following condition $\|u_d(t)\| \leq b_i$ and $\|u_r(t)\| \leq c_i$ with upper bound as $b_i > 0$ and $c_k > 0$ with $i = x, y, z$ and $k = \phi, \theta, \psi$.

Assumption 3 The linear velocities, angular velocities and rate of change of orientation angles v_{s3} , v_{s2} , v_{s3} , Ω_{s1} , Ω_{s2} , Ω_{s3} , $\dot{\phi}$, $\dot{\theta}$ and $\dot{\psi}$ are bounded, respectively.

The proposed adaptive control strategy can be structured into six different controllers. These controllers can be grouped into two loops as internal and external loop. The inner loop consists of roll, pitch, yaw, and altitude controller. The outer or external loop includes position controller. We assume that the external/outer loop runs faster than the inner loop. The position controller generates desired roll and pitch that are tracked by roll and pitch controller. The yaw control moment in attitude controller receives the desired yaw directly from the designer. Altitude controller provides overall thruster for the take-off and landing. Using composite Lyapunov energy function, we explore the control stability and tracking properties of whole closed loop system under proposed algorithms. It is shown in our convergence analysis that the translational and rotational error dynamic con-

verges to zero as the time goes to infinity in Lyapunov sense.

Remark 2 The proposed design assumes that the outer position control loop runs slower than the inner attitude control loop. In practice, this requirement is realistic for quadrotor flying vehicles.

Remark 3 The Euler angles roll and pitch are assumed to be bounded and satisfy the inequalities $\frac{\pi}{2} < \phi < -\frac{\pi}{2}$ and $\frac{\pi}{2} < \theta < -\frac{\pi}{2}$. This assumption is, in fact, practically realizable as the quadrotor vehicle does not perform large maneuver during normal flight condition, such as, vertical landing, vertical take-off and free flight.

Remark 4 It is also important to note that the proposed design require to satisfies the condition for the input $u_1 > 0$ to compensate gravity loading vectors during flying of the vehicle.

3.1 Altitude tracking algorithm and stability analysis

Let us now design altitude tracking system for generating the desired thruster in order to maintain desired distance between the ground and vehicles. The algorithm is assumed that the altitude between the ground and vehicles is available for measurement. The measurement can be obtained using telemetric sensors, such as, barometric pressure, sonar, or laser which gives the actual height of the vehicles. Then, we define the altitude tracking error as

$$e_z = (z_d - z), \quad (15)$$

where z_d is the altitude reference. We define the auxiliary tracking errors signals by combining position and velocity tracking errors as

$$S_z = (\dot{e}_z + \alpha_z e_z), \quad (16)$$

with $\dot{e}_z = (\dot{z}_d - \dot{z})$ and $\alpha_z > 0$. The altitude error dynamics has the following form

$$\ddot{e}_z = \ddot{z}_d - \left(\frac{1}{p_a} (\cos \phi \cos \theta) u_1 - g - Y_{v_{s3}}(v_{s3}) \theta_{v_{s3}} + u_{dz} \right), \quad (17)$$

We then introduce the following altitude tracking control algorithm

$$\begin{aligned}
u_1 &= \frac{p_a}{\cos \phi \cos \theta} (\ddot{z}_d + g + Y_{v_{s3}}(v_{s3})\hat{\theta}_{vs3} + u_{bz} \\
&\quad + \alpha_{z1}\dot{e}_z + k_z S_z), \\
u_{bz} &= -\hat{b}_z \text{sign}(S_z), \\
\dot{\hat{\theta}}_{vs3} &= -\gamma_z Y_{v_{s3}}^T(v_{s3})S_z, \\
\dot{\hat{b}}_z &= \gamma_{pa} \text{sign}^T(S_z)S_z,
\end{aligned} \quad (18)$$

where $\tilde{\theta}_{vs3} = (\hat{\theta}_{vs3} - \theta_{vs3})$, $\tilde{b}_z = (b_z - \hat{b}_z)$, $\gamma_{pa} > 0$, $k_z > 0$, $\alpha_{z1} > \alpha_z$ and $\gamma_z = \text{diag}[\gamma_{z1}, \gamma_{z2}]$. To derive the closed loop stability property, we consider the following Lyapunov function

$$V_z = \frac{1}{2} S_z^T S_z + \frac{1}{2} \tilde{\theta}_{vs3} \gamma_z^{-1} \tilde{\theta}_{vs3} + \frac{1}{2\gamma_{pa}} \tilde{b}_z^2. \quad (19)$$

Taking the time derivative of (19) along the closed loop system constructed by (15)–(18), one can simplify the bound on $\dot{V}_z$ in the following form

$$\dot{V}_z \leq -k_z S_z^T S_z \leq 0, \quad (20)$$

which ensures the bound $e_z \in \mathcal{L}_2$ and $\dot{e}_z \in \mathcal{L}_2$. Then, the boundedness of $V_z \leq V_z(0)$, we have $V_z \in \mathcal{L}_\infty$. Using this fact and assumptions 1 to 2 together with the bound of uncertain parameters, we have $e_z \in \mathcal{L}_\infty$ and $\dot{e}_z \in \mathcal{L}_\infty$. From (15)–(18), it can be easily seen that the signals $\dot{e}_z$ converge to zero if we can show that $\ddot{e}_z$ is bounded and uniformly continuous. In view of assumptions 1 to 3 together with the boundedness of uncertain parameters, the time derivative of $\ddot{e}_z$ is bounded as $\frac{d}{dt}\ddot{e}_z \in \mathcal{L}_\infty$. Thus, $\ddot{e}_z$ is uniformly continuous. Then, using Barbalat's Lemma, we can conclude $\dot{e}_z \rightarrow 0$ as $t \rightarrow \infty$. Since $\dot{e}_z \rightarrow 0$ and $e_z \rightarrow 0$, then $S_z \rightarrow 0$ as $t \rightarrow \infty$. Then, in view of our above analysis, we can conclude that all the tracking error signals in the closed loop system constructed by (15)–(18) are bounded and asymptotically stable.

3.2 Position control algorithm and stability analysis

We now design position control algorithm to generate the desired horizontal motion for the given desired motion trajectories along x and y direction. The horizontal motion can be obtained by rotating the thruster values of the vehicles to the desired direction of motion. The horizontal motion is usually achieved by rolling or pitching the vehicles for the given reference motion x_d and y_d . This means that the position controller generates the desired roll and pitch trajectory to track by attitude controller. To keep the flying vehicle over the

desired point, let us define the commanded acceleration error model along x and y direction as

$$\ddot{e}_x = \ddot{x}_d - \left(\frac{1}{p_a} p_{xi} u_1 - Y_{v_{s1}}(v_{s1})\theta_{vs1} + u_{dx} \right), \quad (21)$$

$$\ddot{e}_y = \ddot{y}_d - \left(\frac{1}{p_a} p_{yi} u_1 - Y_{v_{s2}}(v_{s2})\theta_{vs2} + u_{dy} \right), \quad (22)$$

where p_{xi} and p_{yi} are considered as virtual input. Let us define auxiliary error signals M_x and M_y as

$$M_x = (\dot{e}_x + \alpha_x e_x), \quad (23)$$

$$M_y = (\dot{e}_y + \alpha_y e_y), \quad (24)$$

where $\alpha_x > 0$, $\alpha_y > 0$, $e_x = (x_d - x)$ and $e_y = (y_d - y)$. Then, we design the following virtual position control algorithm in x direction for generating desired rolling motion

$$\begin{aligned}
p_{xi} &= \frac{p_a}{u_1} (\ddot{x}_d + Y_{v_{s1}}(v_{s1})\hat{\theta}_{vs1} + \alpha_{x1}\dot{e}_x + u_{bx} + k_x M_x), \\
\dot{\hat{\theta}}_{vs1} &= -\gamma_x Y_{v_{s1}}^T(v_{s1})M_x, \\
u_{bx} &= -\hat{b}_x \text{sign}(M_x), \\
\dot{\hat{b}}_x &= \gamma_{dx} \text{sign}^T(M_x)M_x,
\end{aligned} \quad (25)$$

where $\gamma_{dx} > 0$, $\tilde{\theta}_{vs1} = (\hat{\theta}_{vs1} - \theta_{vs1})$, $\tilde{b}_x = (b_x - \hat{b}_x)$, $k_x > 0$, $\alpha_{x1} > \alpha_x$ and $\gamma_x = \text{diag}[\gamma_{x1}, \gamma_{x2}]$. The desired pitching motion can be generated using the following virtual position control algorithm in y direction

$$\begin{aligned}
p_{yi} &= \frac{p_a}{u_1} (\ddot{y}_d - Y_{v_{s2}}(v_{s2})\hat{\theta}_{vs2} + \alpha_{y1}\dot{e}_y + u_{by} + k_y M_y), \\
\dot{\hat{\theta}}_{vs2} &= -\gamma_y Y_{v_{s2}}^T(v_{s2})M_y, \\
u_{by} &= -\hat{b}_y \text{sign}(M_y), \\
\dot{\hat{b}}_y &= \gamma_{dy} \text{sign}^T(M_y)M_y,
\end{aligned} \quad (26)$$

where $\gamma_{dy} > 0$, $\tilde{\theta}_{vs2} = (\hat{\theta}_{vs2} - \theta_{vs2})$, $\tilde{b}_y = (b_y - \hat{b}_y)$, $k_y > 0$, $\alpha_{y1} > \alpha_y$ and $\gamma_y = \text{diag}[\gamma_{y1}, \gamma_{y2}]$. For the closed loop system formulated by (21)–(26), we consider the following Lyapunov like energy functional

$$V_x = \frac{1}{2} M_x^T M_x + \frac{1}{2} \tilde{\theta}_{vs1} \gamma_x^{-1} \tilde{\theta}_{vs1} + \frac{1}{2\gamma_{dx}} \tilde{b}_x^2, \quad (27)$$

$$V_y = \frac{1}{2} M_y^T M_y + \frac{1}{2} \tilde{\theta}_{vs2} \gamma_y^{-1} \tilde{\theta}_{vs2} + \frac{1}{2\gamma_{dy}} \tilde{b}_y^2. \quad (28)$$

We now take the derivate (27), (28), and then simplify the time derivative $\dot{V}_x$ and $\dot{V}_y$ along the closed loop trajectories derived by (21)–(26) as follows

$$\dot{V}_x \leq -k_x M_x^T M_x \leq 0, \dot{V}_y \leq -k_y M_y^T M_y \leq 0. \quad (29)$$

This implies that the tracking error signals e_x , e_y , $\dot{e}_x$ and $\dot{e}_y$ are bounded as $e_x \in \mathcal{L}_2$, $e_y \in \mathcal{L}_2$, $\dot{e}_x \in \mathcal{L}_2$ and $\dot{e}_y \in \mathcal{L}_2$. Since the position and velocity tracking errors are bounded, then the composite auxiliary errors are also bounded as $M_x \in \mathcal{L}_2$ and $M_y \in \mathcal{L}_2$. Using this fact, assumptions 1 to 3 and the continuity of $V_x(t) \in \mathcal{L}_\infty$ and $V_y(t) \in \mathcal{L}_\infty$, we can write $e_x \in \mathcal{L}_\infty$, $e_y \in \mathcal{L}_\infty$, $\dot{e}_x \in \mathcal{L}_\infty$, $\dot{e}_y \in \mathcal{L}_\infty$, $M_x \in \mathcal{L}_\infty$ and $M_y \in \mathcal{L}_\infty$. Then, applying Barbalat's Lemma, we can conclude that the signals e_x , e_y , $\dot{e}_x$, $\dot{e}_y$, M_x , and M_y are bounded and their bounds are asymptotically converge to zero in Lyapunov sense.

3.3 Attitude control algorithm design and stability analysis

Let us now design algorithms for stabilizing and tracking the attitude dynamics of the quadrotor flying vehicles in the presence of modeling error and disturbance uncertainties. The attitude tracking system can be seen as the heart of the flying vehicles as they ensures desired orientation values even in the presence of large perturbation from hovering flight to the target point. It is assumed that the orientation angles are available from inertial measurement unit. The tracking errors for the given desired rolling angles can be derived as

$$\ddot{e}_\phi = -[p_b(\dot{\theta}\dot{\phi}) + p_{b1}u_2 - p_{b2}f(u)\dot{\theta} - Y_{\Omega_{s1}}(\Omega_{s1})\theta_{\Omega_{s1}} + u_{r\phi}] + \ddot{\phi}_d, \quad (30)$$

where ϕ_d is the reference rolling angle. Using φ_d , p_x and p_y , ϕ_d can be calculated from the relationship $\phi_d = \arcsin(p_x \sin(\varphi_d) - p_y \cos(\varphi_d))$. Then, we introduce the following control input for generating the desired rolling moment as

$$\begin{aligned} u_2 &= \frac{1}{p_b}[\ddot{\phi}_d - \hat{M}_1 \mathcal{K}_l(\Omega_{s2}, \Omega_{s3}) + \hat{M}_2 \mathcal{M}_l(u, \Omega_{s2}) \\ &\quad + Y_{\Omega_{s1}}(\Omega_{s1})\hat{\theta}_{\Omega_{s1}} + u_{r\phi} + \alpha_{\phi_1}\dot{e}_\phi + k_\phi N_\phi], \\ u_{r\phi} &= -\hat{c}_\phi \text{sign}(N_\phi), \hat{\theta}_{\Omega_{s1}} = -\gamma_\phi Y_{\Omega_{s1}}^T(\Omega_{s1})N_\phi, \\ \dot{\hat{c}}_\phi &= \gamma_{pb1} \text{sign}^T(N_\phi)N_\phi, \dot{\hat{M}}_1 = \gamma_{M1} \mathcal{K}_l^T(\Omega_{s1}, \Omega_{s2})N_\phi, \\ \dot{\hat{M}}_2 &= -\gamma_{M2} \mathcal{M}_l^T(u, \Omega_{s2})N_\phi, \end{aligned} \quad (31)$$

where $k_\phi > 0$, $\mathcal{K}_l(\Omega_{s2}, \Omega_{s3}) = \dot{\theta}\dot{\phi}$, $\mathcal{M}_l(u, \Omega_{s2}) = f(u)\dot{\theta}$, $N_\phi = (\dot{e}_\phi + \alpha_\phi e_\phi)$, $e_\phi = (\phi_d - \phi)$, $\alpha_\phi > 0$, $\alpha_{\phi_1} > \alpha_\phi$, $\gamma_{pb1} > 0$, $\gamma_\phi = \text{diag}[\gamma_{\phi_1}, \gamma_{\phi_2}]$, $\gamma_{M2} > 0$,

$\gamma_{M1} > 0$, $\tilde{\theta}_{\Omega_{s1}} = (\hat{\theta}_{\Omega_{s1}} - \theta_{\Omega_{s1}})$, $\tilde{c}_\phi = (c_\phi - \hat{c}_\phi)$, $\tilde{M}_2 = (\hat{M}_2 - M_2)$ and $\tilde{M}_1 = (M_1 - \hat{M}_1)$. We now choose the following energy function to analyze the closed loop stability under the proposed rolling moment (31)

$$\begin{aligned} V_\phi &= \frac{1}{2} N_\phi^T N_\phi + \frac{1}{2} \tilde{\theta}_{\Omega_{s1}}^T \gamma_\phi^{-1} \tilde{\theta}_{\Omega_{s1}} + \frac{1}{2\gamma_{pb1}} \tilde{c}_\phi^2 + \\ &\quad + \frac{1}{2\gamma_{M1}} \tilde{M}_1^2 + \frac{1}{2\gamma_{M2}} \tilde{M}_2^2. \end{aligned} \quad (32)$$

Taking the time derivative of (32) along the trajectories of (30) with input algorithm (31), we can simplify the derivative of V_ϕ as

$$\dot{V}_\phi \leq -k_\phi N_\phi^T N_\phi \leq 0. \quad (33)$$

Then, using Lyapunov argument, we can write from Eq. (33) as e_ϕ , $\dot{e}_\phi$ and N_ϕ are bounded as $e_\phi \in \mathcal{L}_2$, $\dot{e}_\phi \in \mathcal{L}_2$, $N_\phi \in \mathcal{L}_2$, $e_\phi \in \mathcal{L}_\infty$, $\dot{e}_\phi \in \mathcal{L}_\infty$ and $N_\phi \in \mathcal{L}_\infty$. In view of assumptions 1 to 3 and Barbalat's Lemma, we can conclude that the tracking error signals in the closed loop systems are bounded and their bounds converges to zero as the time goes to infinity.

Let us design tracking control algorithm to generate the desired pitching moment to track the desired pitching angle θ_d . The acceleration errors for the pitching angles can be written as

$$\ddot{e}_\theta = -[p_c(\dot{\phi}\dot{\theta}) + p_{c1}u_3 + p_{c2}f(u)\dot{\phi} + u_{r\theta} - Y_{\Omega_{s2}}(\Omega_{s2})\theta_{\Omega_{s2}}] + \ddot{\theta}_d, \quad (34)$$

By knowing φ_d , ϕ_d , p_x and p_y , θ_d can be generated from the relationship $\theta_d = \arcsin\left(\frac{p_x \sin(\varphi_d) + p_y \cos(\varphi_d)}{\cos(\phi_d)}\right)$. Then, we propose the following pitching control moment to generate the desired pitching motion trajectory

$$\begin{aligned} u_3 &= \frac{1}{p_{c1}} \left[\ddot{\theta}_d - \hat{M}_3 \mathcal{B}_l(\Omega_{s1}, \Omega_{s3}) - \hat{M}_4 \mathcal{C}_l(u, \Omega_{s1}) \right. \\ &\quad \left. + Y_{\Omega_{s2}}(\Omega_{s2})\hat{\theta}_{\Omega_{s2}} + u_{r\theta} + \alpha_{\theta_1}\dot{e}_\theta + k_\theta L_\theta \right], \\ u_{r\theta} &= -\hat{c}_\theta \text{sign}(L_\theta), \hat{\theta}_{\Omega_{s2}} = -\gamma_\theta Y_{\Omega_{s2}}^T(\Omega_{s2})L_\theta, \\ \dot{\hat{c}}_\theta &= \gamma_{pc1} \text{sign}^T(L_\theta)L_\theta, \dot{\hat{M}}_4 = \gamma_{M4} \mathcal{C}_l^T(u, \Omega_{s1})L_\theta, \\ \dot{\hat{M}}_3 &= \gamma_{M3} \mathcal{B}_l^T(\Omega_{s1}, \Omega_{s3})L_\theta, \end{aligned} \quad (35)$$

where $k_\theta > 0$, $\mathcal{C}_l(u, \Omega_{s1}) = f(u)\dot{\phi}$, $\mathcal{B}_l(\Omega_{s1}, \Omega_{s3}) = \dot{\phi}\dot{\theta}$, $L_\theta = (\dot{e}_\theta + \alpha_\theta e_\theta)$, $e_\theta = (\theta_d - \theta)$, θ_d is the reference pitching angle, $\alpha_\theta > 0$, $\alpha_{\theta_1} > \alpha_\theta$, $\gamma_{pc1} > 0$,

Fig. 2 Block diagram representation of the proposed design

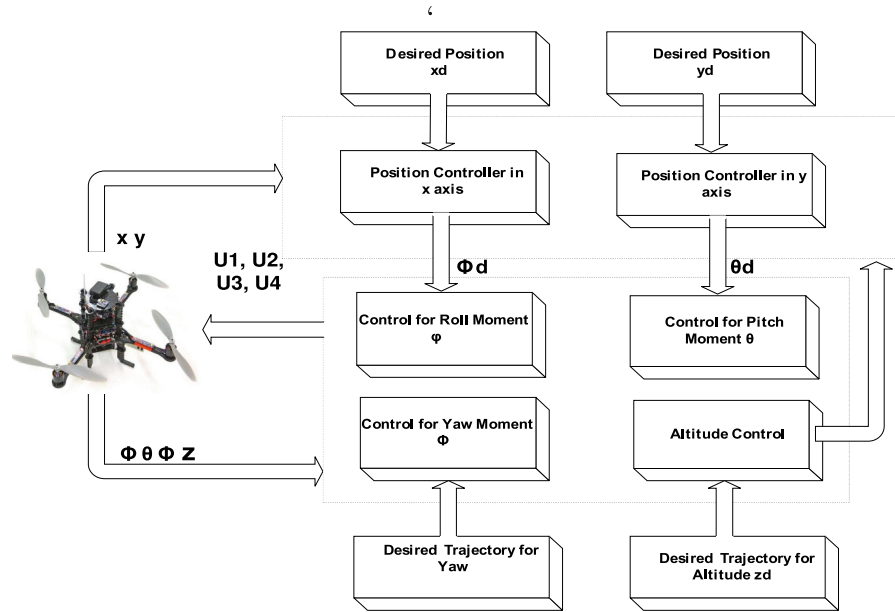


Fig. 3 Commercially available Pelican quadrotor vehicle [18]



$\gamma_\theta \in \mathbb{R}^{2 \times 2}$, $\gamma_{M3} > 0$, $\gamma_{M4} > 0$, $\tilde{\theta}_{\Omega_{s2}} = (\hat{\theta}_{\Omega_{s2}} - \theta_{\Omega_{s2}})$, $\tilde{c}_\theta = (-\hat{c}_\theta + c_\theta)$, $\tilde{M}_3 = (-\hat{M}_3 + M_3)$ and $\tilde{M}_4 = (-\hat{M}_4 + M_4)$. We now use the following Lyapunov energy function for the closed system obtained by (34) under input (35)

$$V_\theta = \frac{1}{2} L_\theta^T L_\theta + \frac{1}{2} \theta_{\Omega_{s2}}^T \gamma_\theta^{-1} \theta_{\Omega_{s2}} + \frac{1}{2\gamma_{pc1}} \tilde{c}_\theta^2 + \frac{1}{2\gamma_{M3}} \tilde{M}_3^2 + \frac{1}{2\gamma_{M4}} \tilde{M}_4^2. \quad (36)$$

Then, the time derivative of (36) can be written in the following simplified form

$$\dot{V}_\theta \leq -k_\theta L_\theta^T L_\theta \leq 0. \quad (37)$$

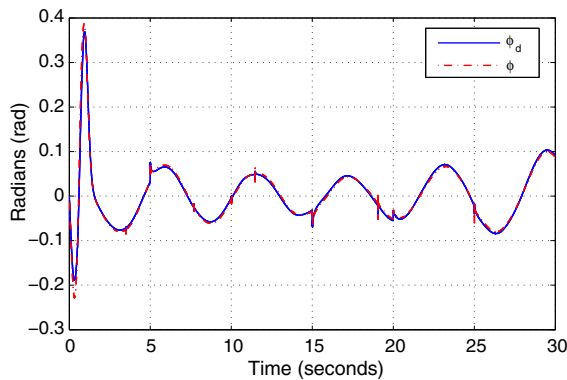
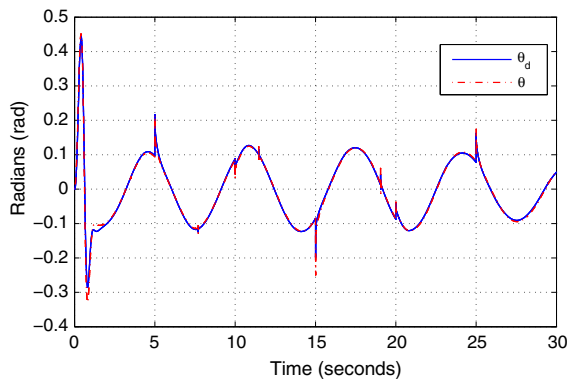
Then, using Lyapunov arguments, we can write $e_\theta \in \mathcal{L}_2$, $\dot{e}_\theta \in \mathcal{L}_2$ and $L_\theta \in \mathcal{L}_2$. Since $e_\theta \in \mathcal{L}_2$, $\dot{e}_\theta \in$

$\mathcal{L}_2$, $L_\theta \in \mathcal{L}_2$ and $V_\theta \in \mathcal{L}_\infty$, we can also obtain the bound on $e_\theta \in \mathcal{L}_\infty$, $\dot{e}_\theta \in \mathcal{L}_\infty$ and $L_\theta \in \mathcal{L}_\infty$. In view of above Lyapunov analysis and assumptions 1 to 3 together with Barbalat's Lemma, we can state that all the error states in the closed loop system (34) under input (35) are asymptotically stable as $\lim_{t \rightarrow \infty} e_\theta = 0$, $\lim_{t \rightarrow \infty} \dot{e}_\theta = 0$ and $\lim_{t \rightarrow \infty} L_\theta = 0$.

Finally, we define angular acceleration tracking error model for the yaw moment as

$$\ddot{e}_\varphi = \ddot{\varphi}_d - [p_d (\dot{\theta} \dot{\phi}) + p_{d1} u_4 + u_{r\psi} - Y_{\Omega_{s3}} (\Omega_{s3}) \theta_{\Omega_{s3}}]. \quad (38)$$

Then, we propose the following control torque to generate the desired yaw moment

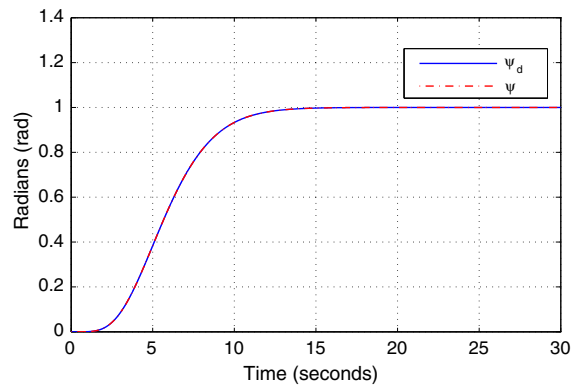

 Fig. 4 Euler angles ϕ_d and ϕ in radians

 Fig. 5 Euler angles θ_d and θ in radians

$$\begin{aligned}
 u_4 &= \frac{1}{p_{d1}} [\ddot{\phi}_d - \hat{M}_5 \mathcal{D}_l(\Omega_{s1}, \Omega_{s2}) + Y_{\Omega_{s3}}(\Omega_{s3}) \hat{\theta}_{\Omega_{s3}} \\
 &\quad + u_r \psi + \alpha_{\varphi_1} \dot{e}_\varphi + k_\varphi P_\varphi], \\
 u_r \psi &= -\hat{c}_\psi \text{sign}(P_\varphi), \quad \dot{\hat{\theta}}_{\Omega_{s3}} = -\gamma_\varphi Y_{\Omega_{s3}}^T(\Omega_{s3}) P_\varphi, \\
 \dot{\hat{c}}_\psi &= \gamma_{pd1} \text{sign}^T(P_\varphi) P_\varphi, \quad \dot{\hat{M}}_5 = \gamma_M \mathcal{D}_l^T(\Omega_{s1}, \Omega_{s2}) P_\varphi,
 \end{aligned} \quad (39)$$

where $k_\varphi > 0$, $\mathcal{D}_l(\Omega_{s1}, \Omega_{s2}) = \dot{\theta}\dot{\phi}$, $P_\varphi = (\dot{e}_\varphi + \alpha_\varphi e_\varphi)$, $e_\varphi = (\varphi_d - \varphi)$, φ_d is the reference yaw angle, $\alpha_\varphi > 0$, $\alpha_{\varphi_1} > \alpha_\varphi$, $\gamma_{pd1} > 0$, $\gamma_M > 0$, $\gamma_\varphi = \text{diag}[\gamma_{\varphi_1}, \gamma_{\varphi_2}]$, $\tilde{\theta}_{\Omega_{s3}} = (\hat{\theta}_{\Omega_{s3}} - \theta_{\Omega_{s3}})$, $\tilde{M}_5 = (M_5 - \hat{M}_5)$ and $\tilde{c}_\psi = (\hat{c}_\psi - c_\psi)$. To guarantee the closed loop stability with the yaw moment provided by (39), we consider the following positive definite function

$$V_\varphi = \frac{1}{2} P_\varphi^T P_\varphi + \frac{1}{2} \theta_{\Omega_{s3}}^T \gamma_\varphi^{-1} \theta_{\Omega_{s3}} + \frac{1}{2\gamma_{pd1}} \tilde{c}_\psi^2 + \frac{1}{2\gamma_{M5}} \tilde{M}_5^2. \quad (40)$$

Take the time derivative (40) along the closed loop tracking error model (38) with input torque (39), we


 Fig. 6 Euler angles ψ_d and ψ in radians

can simplify $\dot{V}_\varphi$ as follows

$$\dot{V}_\varphi \leq -k_\varphi P_\varphi^T P_\varphi \leq 0. \quad (41)$$

In view of Eq. (41), the bound on the error signals e_φ , $\dot{e}_\varphi$ and P_φ can be written as $e_\varphi \in \mathcal{L}_2$, $\dot{e}_\varphi \in \mathcal{L}_2$ and $P_\varphi \in \mathcal{L}_2$. Then, using this fact and the continuity of the V_φ also ensures the bound $e_\varphi \in \mathcal{L}_\infty$, $\dot{e}_\varphi \in \mathcal{L}_\infty$ and $P_\varphi \in \mathcal{L}_\infty$. Now, in view of assumptions 1 to 3, Barbalat's Lemma and above Lyapunov analysis, we can also show that the error states, e_φ , $\dot{e}_\varphi$ and P_φ , in the closed loop systems (38) under input (39) are bounded and their bounds asymptotically converge to zero as the time goes to infinity. The overall representation of the proposed control algorithm is depicted in Fig. 2. Based on our above analysis, we can state our main results in the following Theorem 1.

Theorem 1 *Let assumptions 1 to 3 hold. Then, all the error states in the closed loop tracking error systems formulated by Eqs. (17), (18), (21), (22), (25), (26), (30), (31), (34), (35), (38), (39), are bounded and their bounds asymptotically converge to zero as the time goes to infinity.*

Proof To prove Theorem 1, we consider the following composite Lyapunov function

$$V_c = V_x + V_y + V_z + V_\phi + V_\theta + V_\psi. \quad (42)$$

In view of Eqs. (20), (29), (33), (37), (41), we can simplify $\dot{V}_c$ as

$$\begin{aligned}
 V_c &\leq -k_z S_z^T S_z - k_x M_x^T M_x - k_y M_y^T M_y - k_\phi N_\phi^T N_\phi \\
 &\quad - k_\theta L_\theta^T L_\theta - k_\varphi P_\varphi^T P_\varphi \leq 0.
 \end{aligned} \quad (43)$$

Then, in view of our previous analysis for (20), (29), (33), (37), (41), we can conclude that all the error states

Fig. 7 Position tracking in x , y and z direction in meters

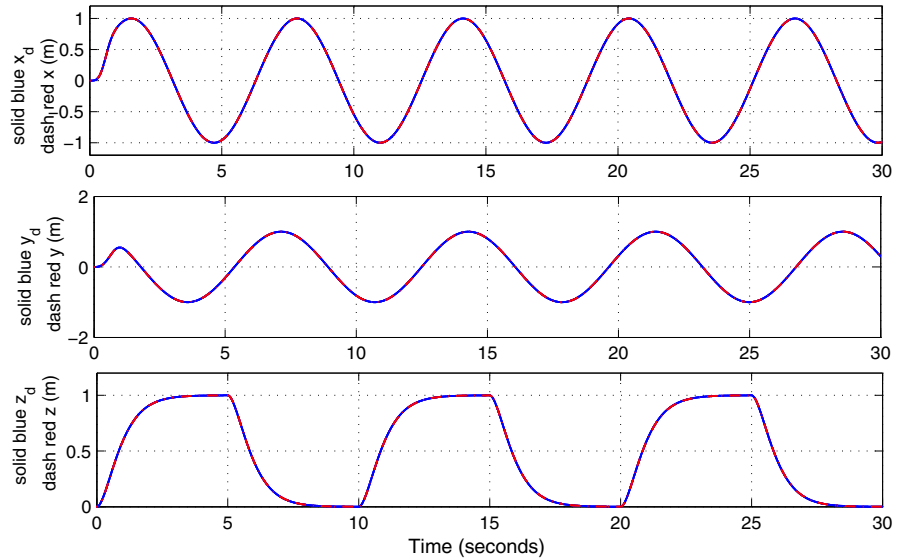
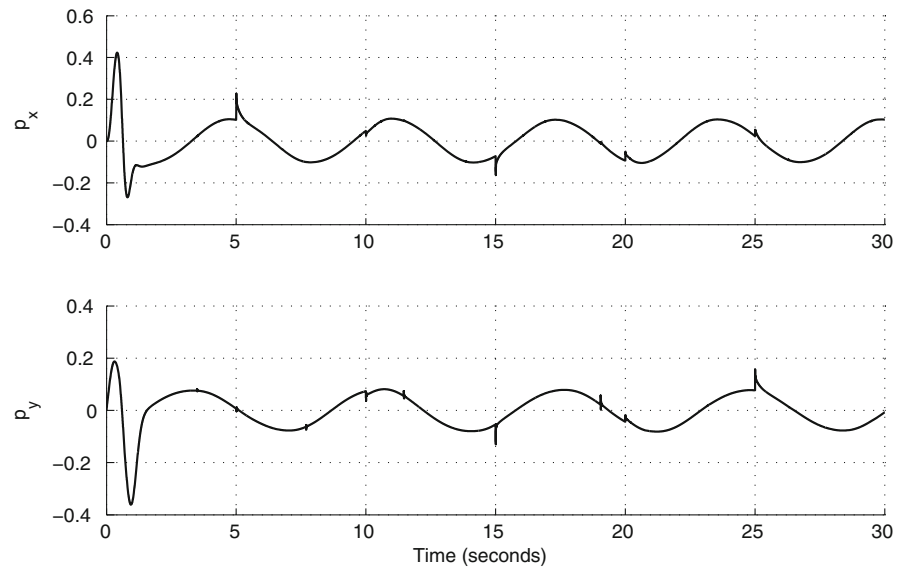


Fig. 8 Virtual input p_x and p_y in Newton-meter



signals in the closed loop systems formulated by Eqs. (17), (18), (21), (22), (25), (26), (30), (31), (34), (35), (38), (39) are bounded and asymptotically stable in Lyapunov sense.

4 Simulation results and experimental demonstration

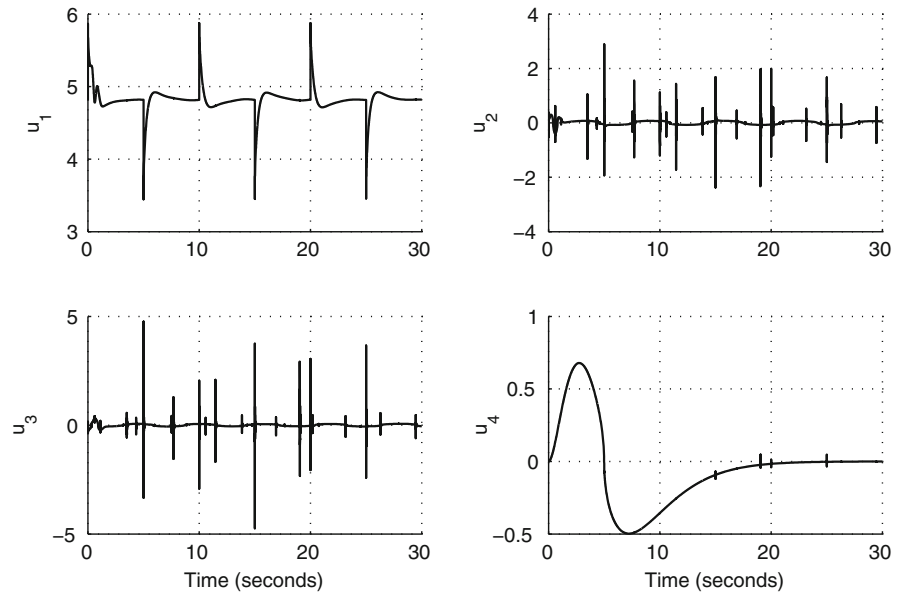
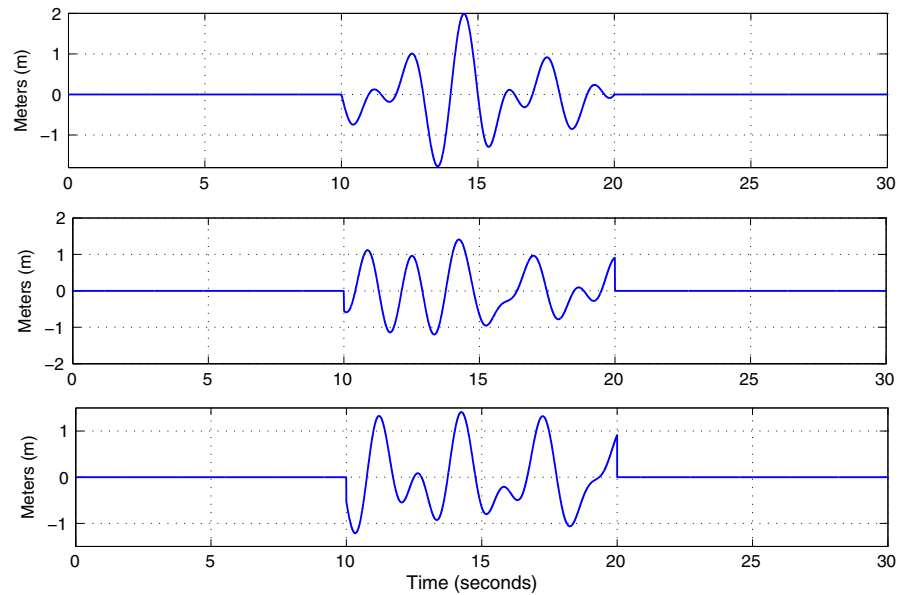
In this section, we first evaluate the proposed algorithms on a simulated model of a commercial Pelican quadrotor UAV provided by Ascending Technologies, GmbH as shown in Fig. 3 [18]. Our goal in this evalua-

tion is to examine the flight control stability and tracking property of the proposed algorithms with respect to varying mass, aerodynamic damping, inertia matrix, and external disturbance uncertainty. For our evaluation, the time varying desired trajectories x_d and y_d are chosen as

$$x_d(t) = (1 - e^{-5t^3}) \sin(t)m,$$

$$y_d(t) = (1 - e^{-2t^3}) \cos(.88t)m,$$

In order to generate desired trajectory for z_d , a square wave with a period of 8 seconds is and an amplitude

Fig. 9 Inputs u_1, u_2, u_3, u_4 in Newton-meter**Fig. 10** State dependant disturbance uncertainty acting along with x, y, z axis

of ± 1 radians is pre filtered with a critically damped 2nd-order linear filter using a bandwidth of $\omega_n = 2.0$ rad/sec. The transfer function for this filter can be described as

$$H(s) = \frac{4}{s^2 + 4s + 4}, \quad (44)$$

with Laplace variable s .

The physical parameters of the Pelican quadrotor UAV are given as $m = 0.485$ kg, $l = 0.35$ m,

$d = 2.9842 \times 10^{-5}$ N/rad/s, $g = 9.81$ m s $^{-2}$, $\alpha_r = 3.2320$ N m/rad/s, $I_x = 3.4 \times 10^{-3}$ kg m 2 , $I_y = 3.4 \times 10^{-3}$ kg m 2 , $I_z = 4.7 \times 10^{-3}$ kg m 2 and $I_r = 3.4 \times 10^{-5}$ kg m 2 . The damping coefficients are chosen as $\zeta_1 = 0.002$ N/m/s, $\zeta_2 = 0.005$ N/m/s, $\zeta_3 = 0.006$ N/m/s, $\zeta_4 = 0.006$ N/m/s, $\zeta_5 = 0.007$ N/m/s, $\zeta_6 = 0.003$ N/m/s, $\xi_1 = 0.001$ N/rad/s, $\xi_2 = 0.003$ N/rad/s, $\xi_3 = 0.006$ N/rad/s, $\xi_4 = 0.006$ N/rad/s, $\xi_5 = 0.008$ N/rad/s and $\xi_6 = 0.003$ N/rad/s.

Fig. 11 State dependent disturbance uncertainty included in ϕ , θ , ψ axis

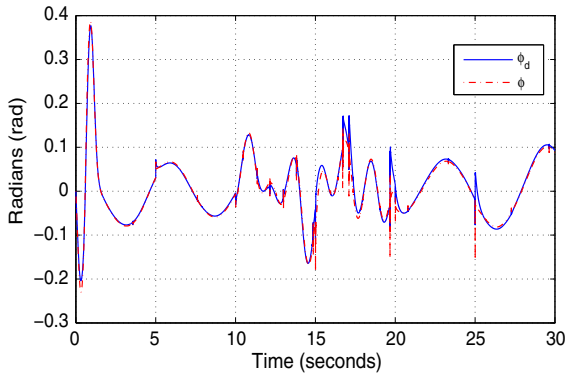
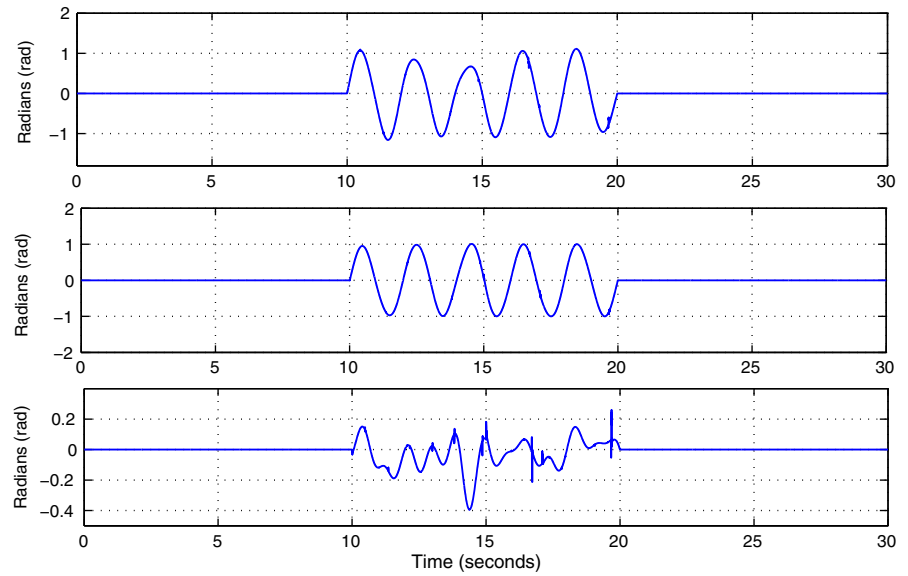


Fig. 12 Euler angles ϕ_d and ϕ in radians

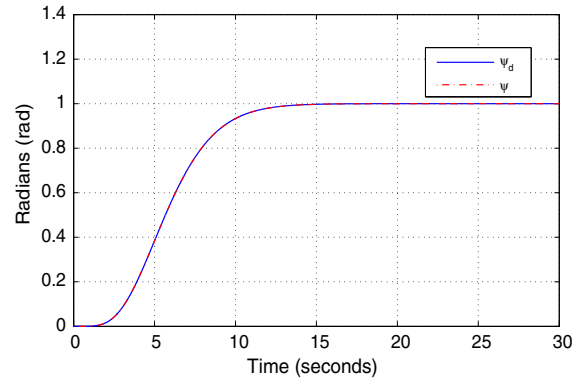


Fig. 14 Euler angles ψ_d and ψ in radians

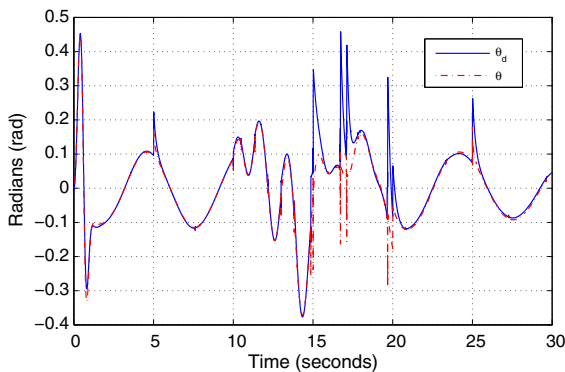


Fig. 13 Euler angles θ_d and θ in radians

The control design parameters are chosen arbitrarily as $\gamma_{M1} = 1$, $\gamma_{M2} = 1$, $\gamma_{M3} = 1$, $\gamma_{M4} = 1$, $\gamma_{M5} = 1$, $\gamma_x = [1 \ 0; 0 \ 1]$, $\gamma_y = [1 \ 0; 0 \ 1]$, $\gamma_z = [1 \ 0; 0 \ 1]$,

$\gamma_{dx} = 1$, $\gamma_{dy} = 1$, $\gamma_{pa} = 2$, $\gamma_{pb1} = 2$, $\gamma_{pc1} = 2$, $\gamma_{pd1} = 2$, $\alpha_x = 2$, $\alpha_y = 2$, $\alpha_z = 2$, $\alpha_\phi = 5$, $\alpha_\theta = 5$, $\alpha_\psi = 5$, $k_x = 200$, $k_y = 200$, $k_z = 100$, $k_\phi = 600$, $k_\theta = 600$, $k_\psi = 100$, $\alpha_{x1} = 80$, $\alpha_{y1} = 80$, $\alpha_{z1} = 80$, $\alpha_{\phi1} = 300$, $\alpha_{\theta1} = 300$ and $\alpha_{\psi1} = 150$. Using these design parameter set-up, we first apply the proposed design on the given unmanned vehicle by assuming that the vehicle is free from uncertainty. The simulation results are given in Figs. 4, 5, 6, 7, 8 and 9. From these results, we can see that the position, altitude, and attitude of the vehicle converge to the reference position, attitude, and altitude with very small errors.

Let us now increase the model parameters in order to increase the modeling error uncertainties. In this evaluation, we increase the mass, inertia, and non-

Fig. 15 Position tracking in x , y and z direction in meters

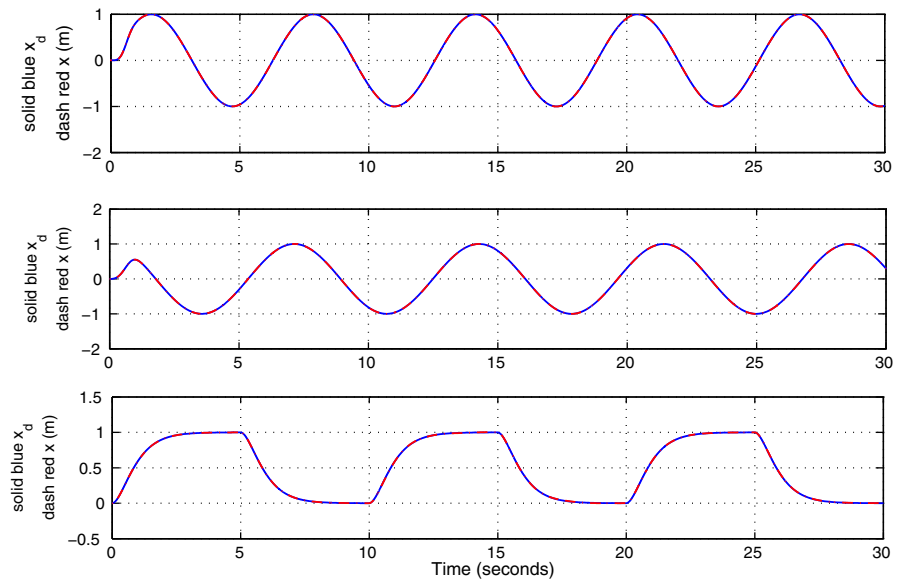
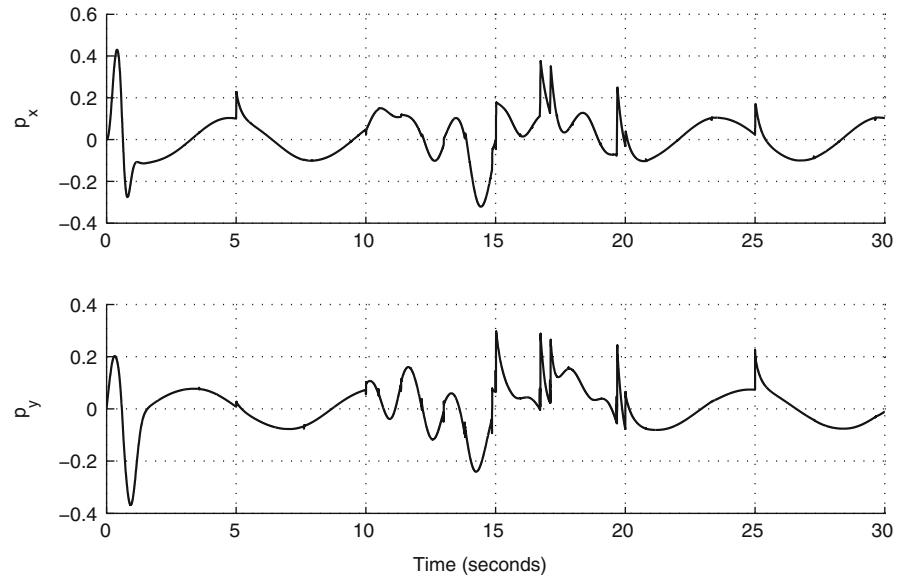


Fig. 16 Virtual inputs p_x and p_y in Newton-meter



linear aerodynamic damping parameters of the vehicle as $m = 5.485$ kg, $I_x = 5.4 \times 10^{-3}$ kg m², $I_y = 5.4 \times 10^{-3}$ kg m², $I_z = 5.7 \times 10^{-3}$ kg m², $I_r = 5.4 \times 10^{-5}$ kg m², $\zeta_1 = 0.2$ N/m/s, $\zeta_2 = 0.5$ N/m/s, $\zeta_3 = 0.6$ N/m/s, $\zeta_4 = 0.6$ N/m/s, $\zeta_5 = 0.7$ N/m/s, $\zeta_6 = 0.3$ N/m/s, $\xi_1 = 0.011$ N/rad/s, $\xi_2 = 0.03$ N/rad/s, $\xi_3 = 0.06$ N/rad/s, $\xi_4 = 0.6$ N/rad/s, $\xi_5 = 0.08$ N/rad/s and $\xi_6 = 0.03$ N/rad/s. We also introduce state dependent external disturbances uncertainties along the translational and rotational dynamics to examine the stability and tracking performance of the proposed design against uncertain flying condition. For our evaluation,

we choose the following state dependant disturbance uncertainties

$$\begin{aligned} u_{dx} &= y \sin(\text{pit}) + z \sin(\text{pit}), u_{dy} = z \sin(\text{pit}) + x \cos(\text{pit}), \\ u_{dz} &= y \sin(\text{pit}) + x \cos(\text{pit}), u_{rx} = \theta \sin(2\text{pit}) + \psi \sin(2\text{pit}), \\ u_{ry} &= \phi \cos(\text{pit}) + \psi \sin(\text{pit}), u_{rz} = \phi \cos(\text{pit}) + \theta \sin(\text{pit}). \end{aligned} \quad (45)$$

Using these design parameters, we then apply the proposed design on the given flying vehicle with and without external disturbance uncertainty. At first, from 0 to 10 s, the vehicle operates under normal condition. Then, from 10 to 20 s, the vehicle flies in the presence

Fig. 17 Inputs u_1, u_2, u_3, u_4 in Newton-meter

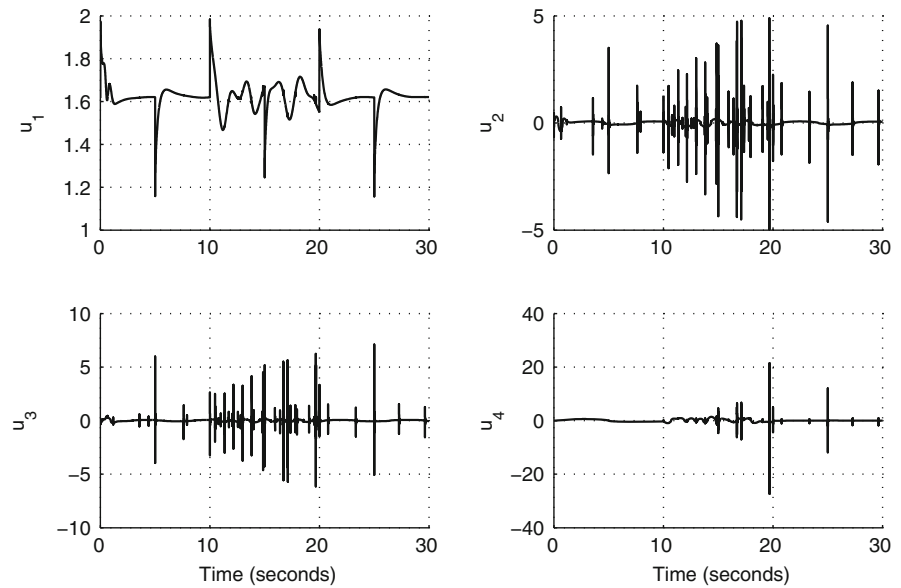
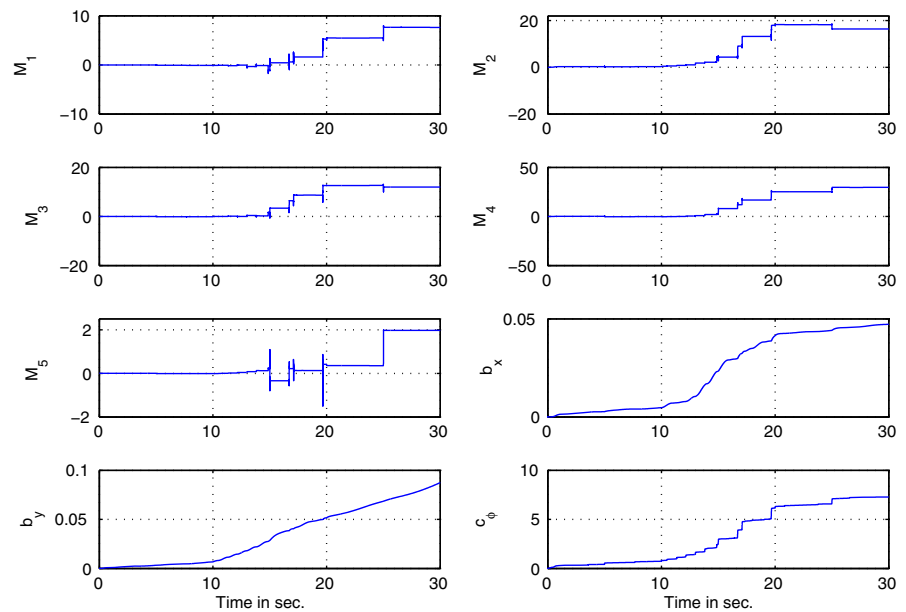


Fig. 18 The time history of the adaptation process $M_1, M_2, M_3, M_4, M_5, b_x, b_y,$ and c_ϕ



of the state dependant uncertainties that entering into translational and rotational axes as shown in Figs. 12, 13, 14, 15, 16 and 17. Finally, from 10 to 20 s, the system returns to normal operating condition without using disturbance uncertainty. The time history of these disturbances is shown in Figs. 10 and 11. The evaluation results are presented in Figs. 12, 13, 14, 15, 16, 17, 18, 19, 20 and 21. For fair comparison, we kept the control and all other design parameters, desired position, and altitude trajectories similar to our previous evalu-

ation. Note that the learning gains and control design parameters can be increased to obtain faster convergence of the tracking error and adaptation process of the parameters. In view of these results, we can see that the tracking error remains closed to zero even with the presence of large modeling error uncertainties.

The experimental evaluation of the proposed design is currently underway on a commercial Hummingbird quadrotor UAV system provided by Asctec Technologies [18]. The preliminary experimental demonstration

Fig. 19 Adaptation process of c_θ , c_ψ , θ_{vs11} , θ_{vs12} , θ_{vs21} , θ_{vs22} , $\theta_{\Omega s11}$ and $\theta_{\Omega s12}$

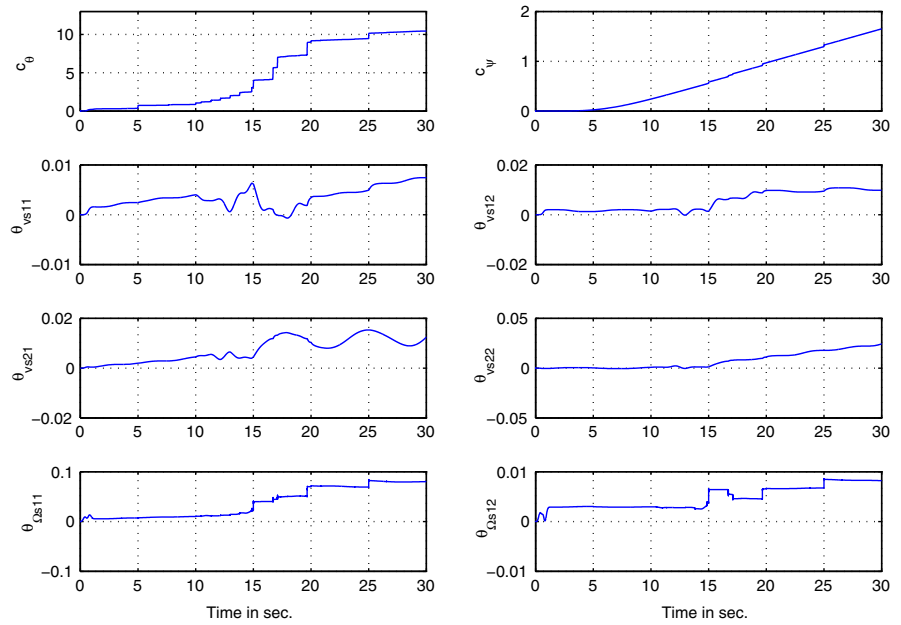


Fig. 20 The time history of the adaptation process $\theta_{\Omega s21}$, $\theta_{\Omega s22}$, $\theta_{\Omega s31}$ and $\theta_{\Omega s32}$

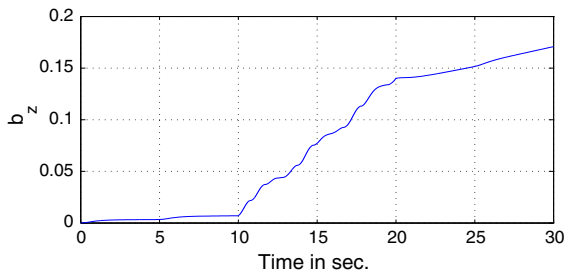
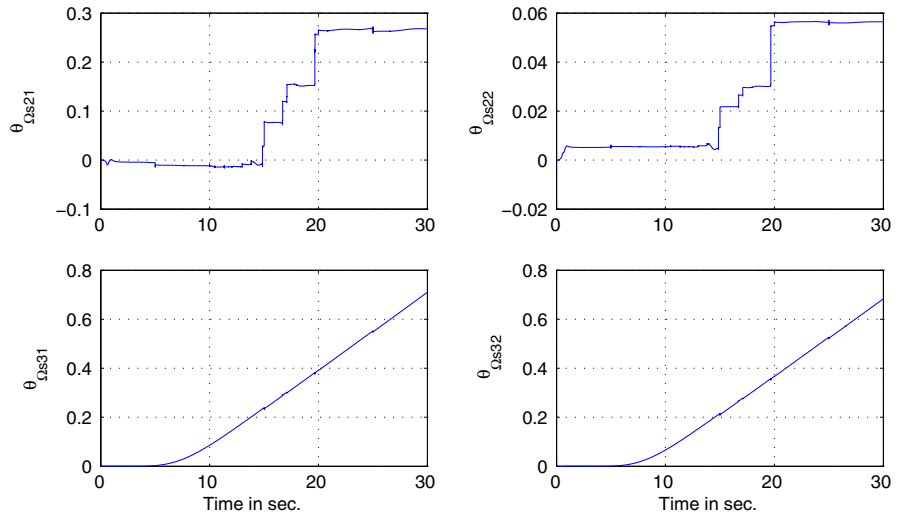


Fig. 21 Adaptation process of b_z

on Hummingbird helicopter is submitted along with the manuscript. These videos can be watched by clicking the website at <http://youtu.be/rOt7QD3i-RA>, <http://youtu.be/X8t9IfbmACI>, <http://youtu.be/mNBmHXBQKZU>, <http://youtu.be/X8t9IfbmACI>, <http://youtu.be/zJGo-9XNFuI>. From these demos, we can see that the proposed method can ensure desired stability and tracking property around the set position with desired height in the presence of the strong disturbances in x , y , z , ϕ , θ and ψ direction.

Remark 5 Note that the adaptation laws proposed in this work may exhibit discontinuous property even after the learning estimate converges to the actual parameter [19, 20]. To ensure that the parameter estimates remain bounded over the given compact sets, one may employ a projection mechanism to limit the learning parameter estimates [20]. To ensure that the adaptation process of the parameters remain bounded over the given compact sets, the projection of the parameter adaptation laws can be written as

$$[\text{Proj}(\hat{\theta}_k, \Phi_k)]_i = \begin{cases} \Gamma_k \Phi_{ki} & \text{if } x_{ki} \leq \hat{\theta}_{ki} \leq y_{ki} \text{ or} \\ & \text{if } \hat{\theta}_{ki} > y_{ki} \text{ and } \Phi_{ki} \leq 0 \text{ or} \\ & \text{if } \hat{\theta}_{ki} < x_{ki} \text{ and } \Phi_{ki} \geq 0 \\ \Gamma_k \bar{\Phi}_{ki} & \text{if } \hat{\theta}_{ki} > y_{ki} \text{ and } \Phi_{ki} > 0 \\ \Gamma_k \check{\Phi}_{ki} & \text{if } \hat{\theta}_{ki} < x_{ki} \text{ and } \Phi_{ki} < 0, \end{cases}$$

where $\bar{\Phi}_{ki} = \left[1 + \frac{y_{ki} - \theta_{ki}}{\delta_k}\right] \Phi_{ki}$, $\check{\Phi}_{ki} = \left[1 + \frac{\theta_{ki} - x_{ki}}{\delta_k}\right] \Phi_{ki}$ and Φ_{ki} is the i -th element of the regression matrices.

Remark 6 Unlike the reported flight control design for quadrotor UAV system, the proposed adaptive method does not require *a priori* known value of the upper bound of the modeling errors and disturbances. The upper bound of the modeling errors and disturbances can be calculated by adaptive laws.

5 Conclusion

In this paper, we have presented nonlinear adaptive tracking system for quadrotor flying vehicles in the presence of the modeling errors and disturbance uncertainties. The algorithms have been developed using Lyapunov energy function by assuming that all the states are available for measurement. The design can be used to reject the effects of the modeling errors and disturbances acting on the four rotor flying vehicles while maintain the bounded stability and tracking control property of the whole closed loop system. Unlike existing adaptive design, the proposed algorithm does not require *a priori* known bound of the modeling errors and disturbance uncertainty. The bound can be obtained by designing an adaptive law. Therefore, the proposed method can be applied to quadrotor system with large modeling error and disturbance uncertainty associated with payload mass, external disturbances associated with uncertain indoor and outdoor environment, such as, strong wind gust, moment of inertia, nonlinear aerodynamic friction, and gyroscopic effects. Simulation results and experimental demonstration on

an unmanned commercial quadrotor vehicle have presented to demonstrate the theoretical development of this paper.

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